

Surname	Centre Number	Candidate Number
Other Names		2

GCE A LEVEL



1400U30–1

BIOLOGY – A2 unit 3

Energy, Homeostasis and the Environment

2 hours

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	14	
2.	15	
3.	8	
4.	13	
5.	14	
6.	17	
7.	9	
Total	90	

ADDITIONAL MATERIALS

In addition to this paper, you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question 7.

The quality of written communication will affect the awarding of marks.

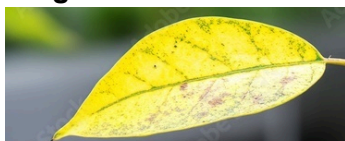
1. Photosynthesis is an important process which occurs in plants.

- (a) Name the elements required for protein synthesis and, beside protein synthesis, explain the role of elements in plants.

[2]

- (b) **Image 1.1** shows an abnormal leaf which is mainly yellow in colour.

Image 1.1



Nematode infection is when plant could not absorb iron and manganese ions from the soil even if it is highly concentrated.

Both manganese and iron ions have a similar role compared to magnesium in the leaf.

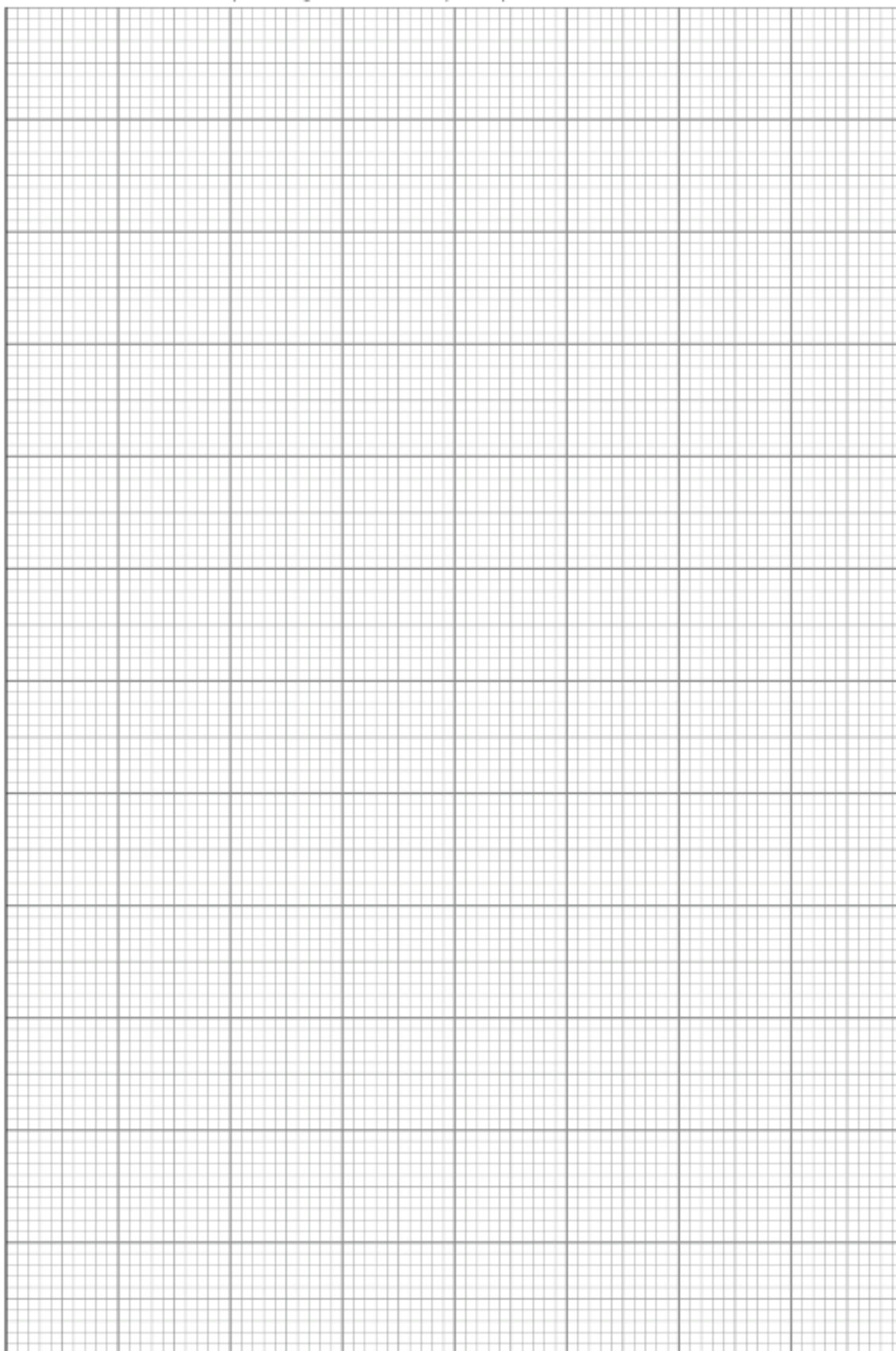
Explain the role of manganese ions and why nematode infection plants turn yellow in their leaf.

[3]

- (c) An investigation was done by Ian where the effect of minerals was investigated.
The data for the experiment was recorded in the table below.

Table 1.2

Concentration of Sach's water culture solution (mol dm ⁻³)	Dry mass of the germinated seedlings (g)			
	Trial 1	Trial 2	Trial 3	Mean
10.0	5.00	6.00	5.00	
20.0	10.00	12.00	13.00	
30.0	17.00	18.00	16.00	
40.0	18.00	19.00	19.00	
50.0	20.00	19.00	20.00	
60.0	22.00	22.00	22.00	
70.0	22.00	22.00	22.00	



(i) Calculate the mean mass for each of the concentration and fill it in **Table 1.2** [1]

(ii) Draw a bar chart for the dry mass of the germinated seedlings using the data included in **Table 1.2** [5]

(iii) Describe the trend graph and explain the effect of concentration of Sach's water culture solution on the dry mass.

[3]

2. This question is about human impact on the environment and planetary boundaries.

(a) (i) State two planetary boundaries which have been crossed.

[1]

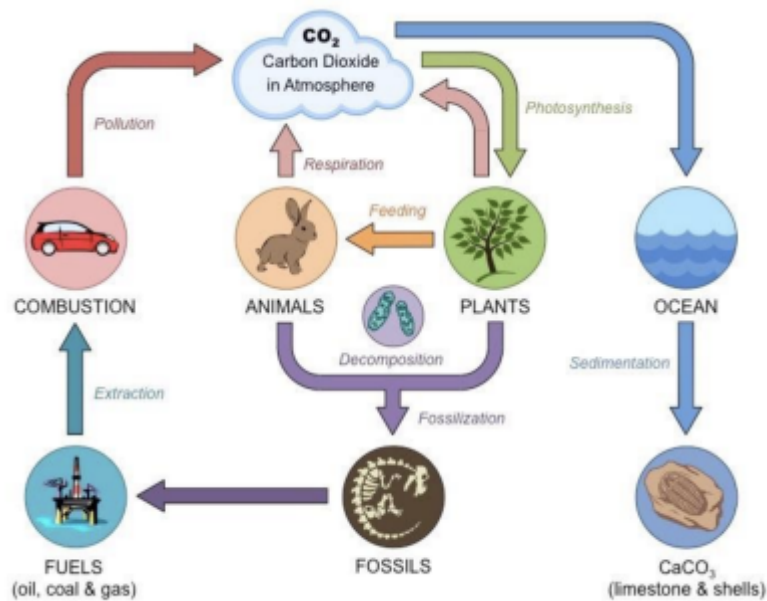
_____ and _____

(ii) Explain the consequences for one of the boundaries which has been crossed.

[2]

(b) **Image 2.1** shows an Carbon Cycle

Image 2.1



(i) Label clearly, using an arrow, on **Image 2.1** where photoautotrophs and saprotrophs are present in the Carbon Cycle.

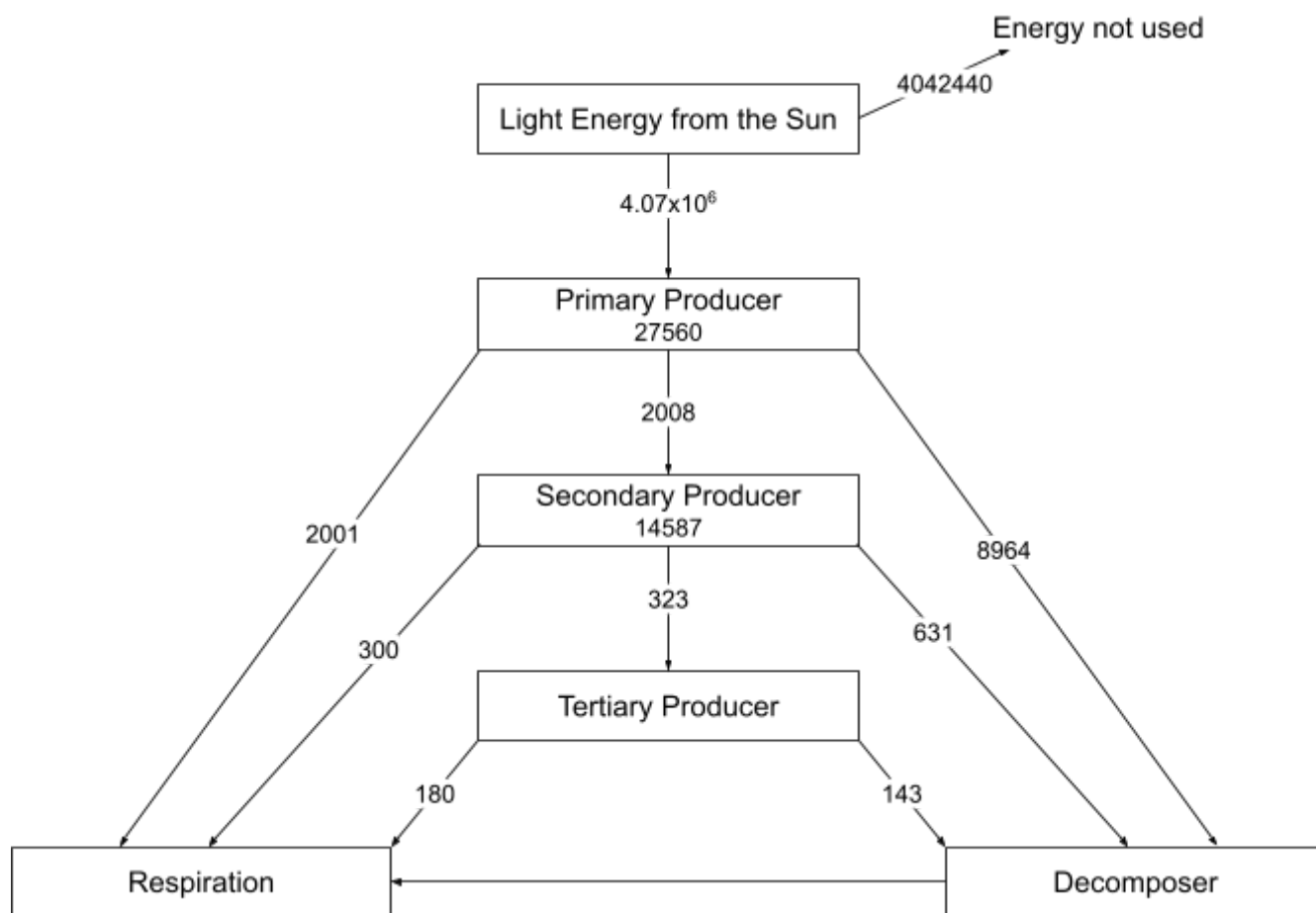
[2]

(ii) Explain in terms of photoautotrophs and saprotrophs' role in the Carbon Cycle

[2]

- (c) Multiple food chains will form a food web and it is commonly found in a habitat.
Diagram 2.2 shows a food chain ecosystem in Cardiff, the energy are shown in $\text{kJ m}^{-2} \text{ year}^{-1}$

Diagram 2.2



- (i) Explain what the arrows mean in the food chain

[1]

- (ii) Aphide is a tertiary producer, using the information in **Diagram 2.2** and the equation below, calculate the mass of Aphide, giving your answer to three significant figures. Show your working in full.

[2]

$$\text{Mass of Organism} = \text{Energy transferred to the organism} \times 2\pi$$

Mass of Aphids = _____ kg

- (d) (i) Fruit and vegetables may contain residues of pesticides, such as **acetylcholinesterase** inhibitors, or heavy metal ions, which can be toxic if ingested. Many of these substances act as neurotoxins because they interfere with synaptic transmission.

Explain in terms of synaptic transmission, how does washing fruit and vegetables to avoid humans ingesting toxic levels.

[3]

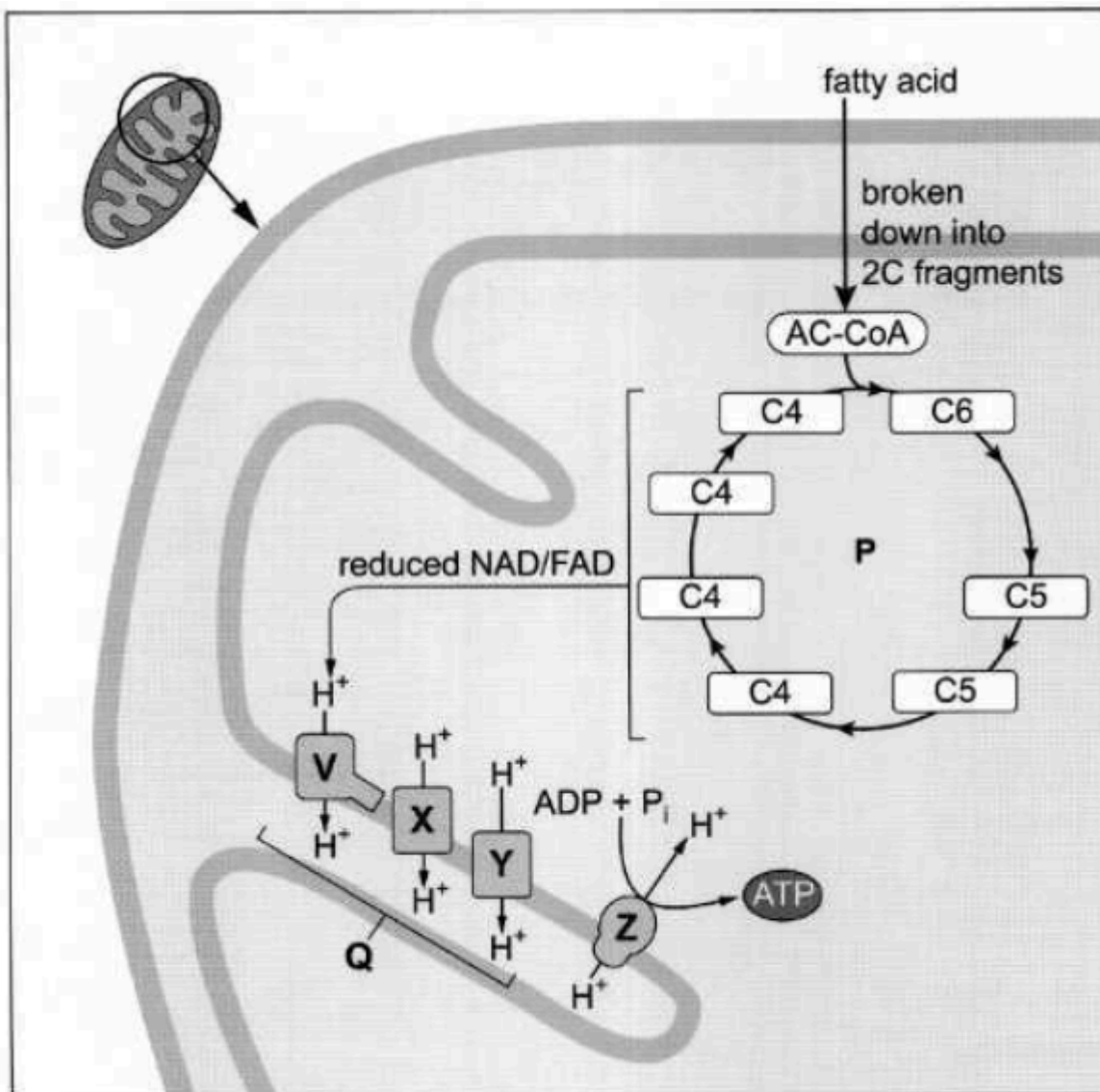
- (ii) Acetylcholinesterase is also used in biosensors to monitor blood glucose concentration.

State **two** advantages of using immobilised enzymes compared to using free enzymes in biosensor.

[2]

3. In the digestion of lipids, fatty acid becomes available for an energy source.
Image 3.1 shows how fatty acids are produced in cells.

Image 3.1



(a) Give the answers for the following question using **Image 3.1**

[2]

The organelle where fatty acid metabolism takes place;

The pathway labelled P;

The pathway labelled Q;

The enzyme contained in the structure labelled Z

- (b) During fatty acid metabolism, fatty acids are broken down into two-carbon fragments. Each two-carbon fragment is converted into one molecule of acetyl-CoA, which then enters pathway P.

State how many molecules of the following compounds will be generated from one molecule of acetyl CoA directly from pathway P. [1]

Reduced NAD _____

Reduced FAD _____

ATP _____

- (c) (i) Describe what is represented by the arrows on the following structures in **Image 3.1**.

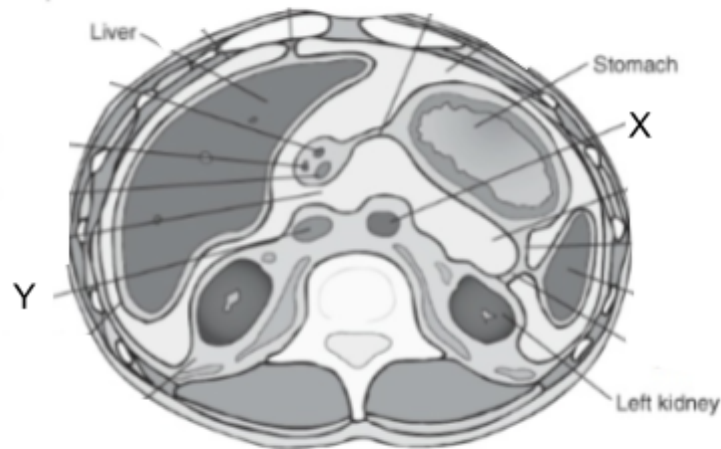
I. Proteins: V, X and Y; [2]

II. Structure Z [2]

- (ii) Water is formed in pathway Q
State the role of oxygen in the pathway. [1]

4. **Image 4.1B** shows a section through the human abdomen.

Image 4.1A



(a) Answer the following questions about the section through the abdomen shown in **Image 4.1A**

- I. Identify the plane of the section shown in **Image 4.1A** [1]

- II. Name the two blood vessels labelled in **Image 4.1A** [2]

- III. State which of the regions of the kidney, labelled X or Y on **Image 4.1B** is the site of ultrafiltration. [1]

The filtration barrier in a nephron has three components. **Image 4.2** shows a diagrammatic representation of the three components. The values given represent the effective pore size of each component.

Image 4.2

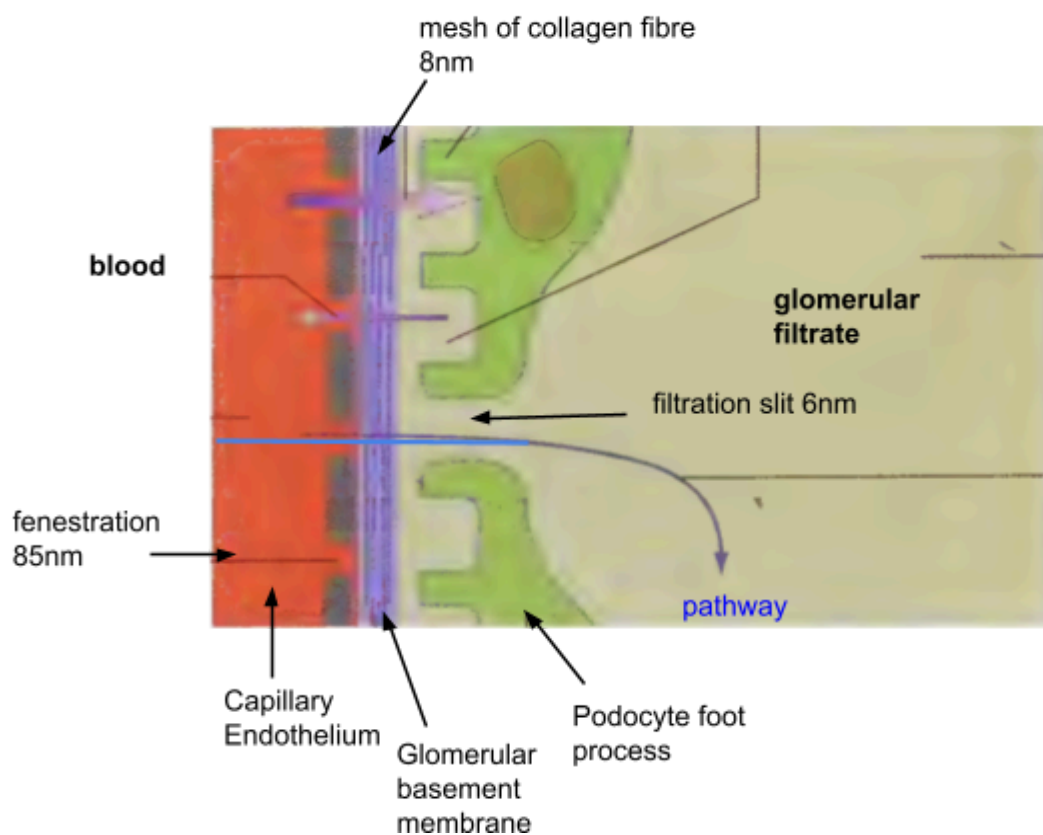


Table 4.3 shows the diameters of some components of blood.

Table 4.3

Component	Diameter / nm
red blood cells	7000
albumin (plasma protein)	7.1
glucose	1.0

(b) With reference to **Image 4.2** and **Table 4.3**, complete **Table 4.4** to explain the presence or absence of albumin and glucose in the glomerular filtrate.

[2]

Table 4.4

Component of blood	Present in glomerular filtrate	Reasons
Album	no	
Glucose	yes	

Question continued overleaf

(c) (i)

The total length of the capillaries in a single glomerulus is 70mm, and one kidney contains one million glomeruli. The mean radius of glomerular capillaries is 0.002 mm.

Use the formula below to calculate the total area of the capillary walls in one kidney. Give your answer in standard form. [3]

Area of capillary walls in one kidney = $2 \pi r l n$.

where

$\pi = 3.14$,

r = radius,

l = length of capillaries in a single glomerulus,

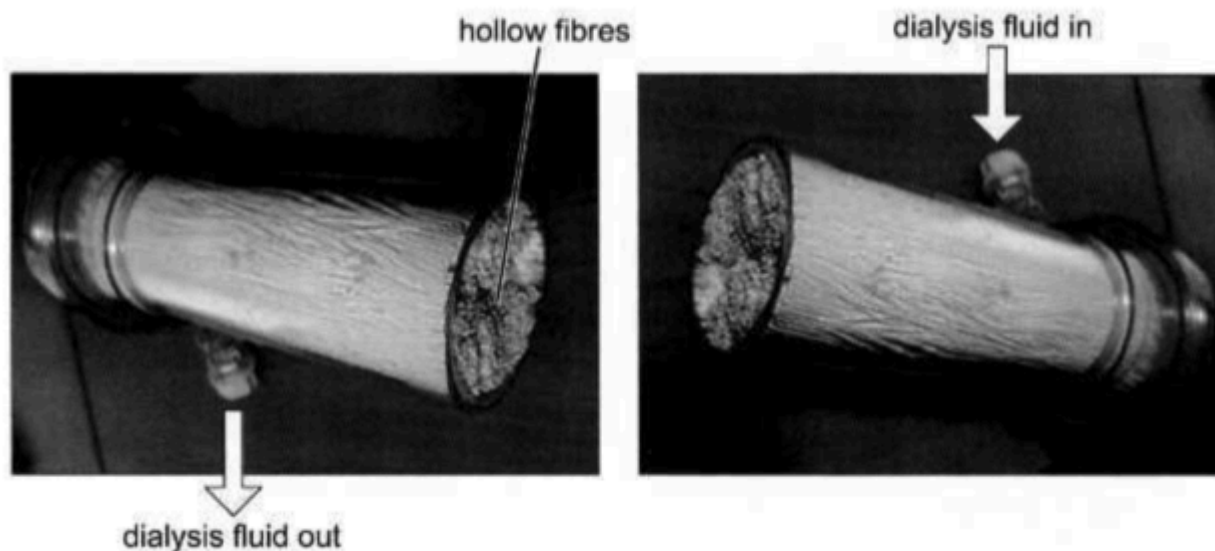
n = the number of glomeruli in a kidney

Total area of the capillary walls in one kidney = _____ mm²

Image 4.5 shows a part of a filter from a dialysis machine cut in half. It contains about 10000 hollow fibres which form the filtration membrane.

The estimated total area of the filtration membrane in the dialysis machine is 2.2m² (1m² = 1 × 10⁶ mm²).

Image 4.5



(ii) Give **one** reason why the estimate of the area for the dialysis filter is likely to be more accurate than the calculated area for the capillary walls of the kidney [1]

(iii) Explain why having a large surface area is important in ultrafiltration [1]

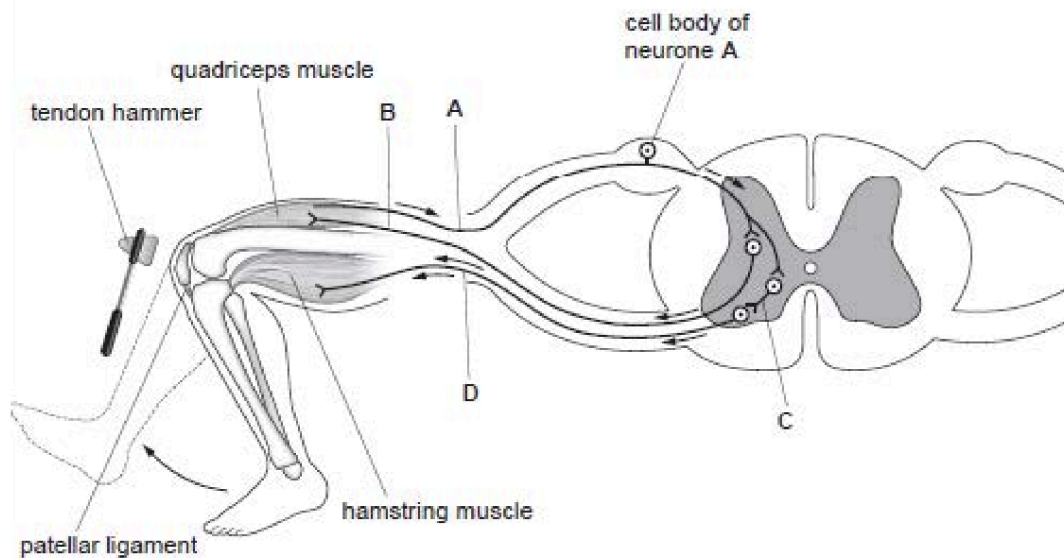
(d) Alport syndrome is a genetic condition that results in the thickening of the basement membrane in the glomeruli. It is caused by an X-linked recessive allele.

(i) Suggest how the filtration rate would be affected in Alport syndrome.
Explain your answer. [1]

(ii) State what is meant by the term X-linked
Explain why males are more likely to have Alport syndrome than females. [2]

5. Reflex actions are defined as being rapid and automatic. **Image 5.1** shows the reflex arc involved in the knee-jerk reflex.

Image 5.1



- (a) (i) Use **Image 5.1** to name the effector and the part of the Central Nervous System (CNS) involved in this reflex. [1]

Effector _____

Part of CNS _____

- (ii) Name the types of neurons labelled **A** and **D**. [1]

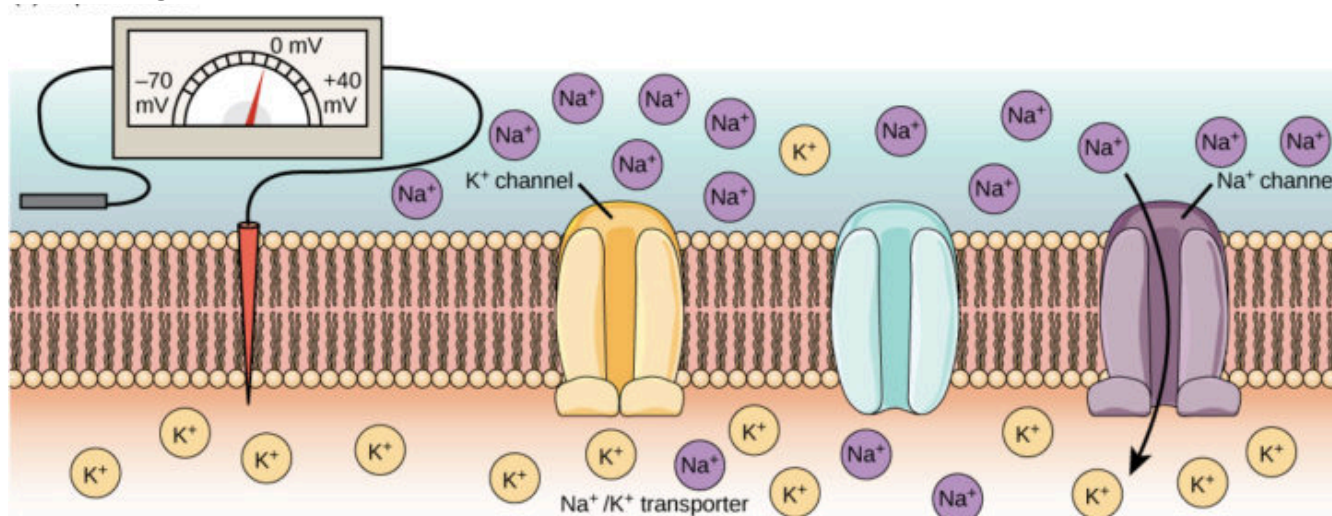
A _____

B _____

- (iii) Explain, in terms of the structures visible in **Image 5.1**, why this reflex is **rapid** and **automatic**. [2]

- (b) **Image 5.2** shows a stage in the generation of an action potential across the cell membrane of a neuron.

Image 5.2

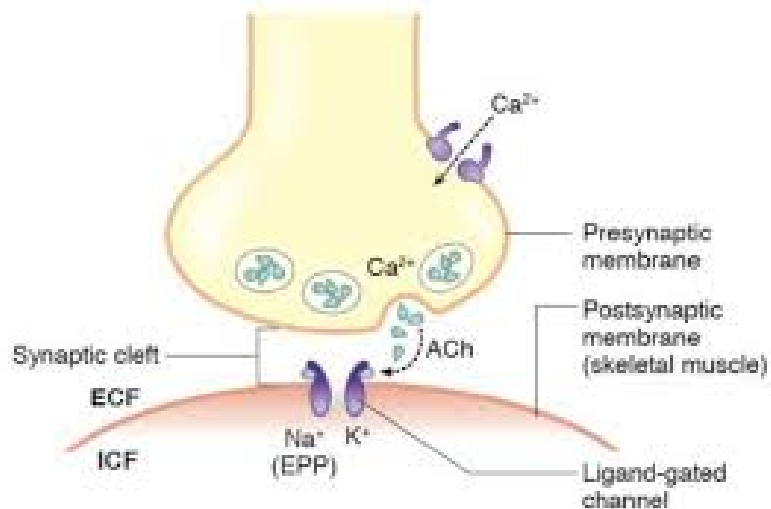


- (i) Name the stage of the action potential shown in **Image 5.2** [1]
- _____
- (ii) Describe how you would expect the needle on the microvoltmeter to move as this stage progresses [2]
- _____
- (iii) Explain the roles of potassium ion channels and sodium ion channels in the generation of action potential [2]
- _____
- _____
- _____
- _____
- (iv) Explain the role of sodium-potassium ion pump in returning the cell membrane of the neurone to its resting potential [2]
- _____
- _____
- _____
- _____

(c) Synapses between neurons and muscles are called neuromuscular junctions. When a nerve impulse reaches a neuromuscular junction, it triggers the release of acetylcholine (ACh) as in many other synapses.

Lambert Eaton Syndrome (LES) is an autoimmune disease in which a person produces antibodies that block some calcium ion channels in the presynaptic membranes of neuromuscular junctions as shown in **Image 5.3**

Image 5.3



To diagnose LES, doctors test for knee-jerk reflexes.

Explain why someone with LES has no knee-jerk reflexes

[4]

6. Bacterial resistance to antibiotics is a serious concern in modern medicine so a great deal of research is being carried out to discover new antibiotics.

- (a) Microbiologists tested the effectiveness of four antibiotics (A–D) by measuring their Minimum Inhibitory Concentration (MIC). This is the lowest concentration of an antibiotic that will inhibit the visible growth of a microorganism. This is done using a microdilution plate, as shown in **Image 6.1A**. **Image 6.1B** shows a side view of the microdilution plate and the concentrations of antibiotics produced in the different wells.

Image 6.1A

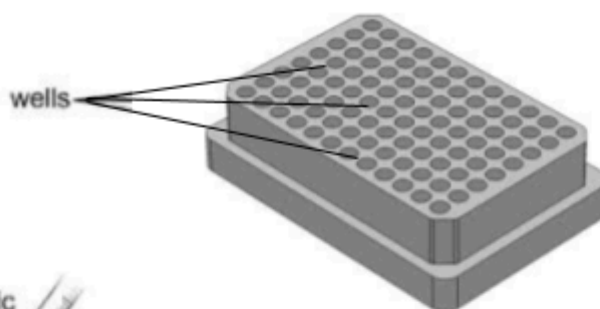
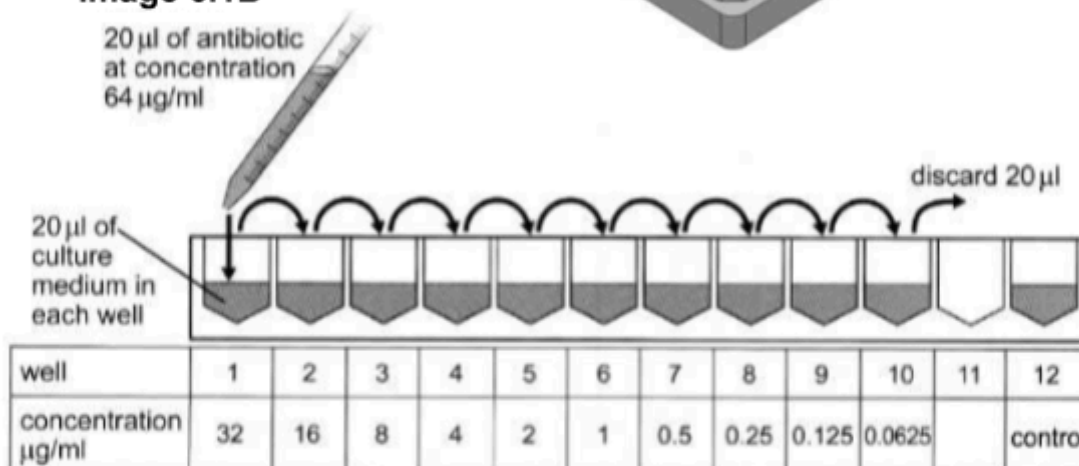


Image 6.1B



The method used is shown below:

1. Place 20 μl of culture medium in wells 1–10 of a microdilution plate.
2. Add 20 μl of antibiotics at concentration 64 $\mu\text{g/ml}$ to well 1.
3. Transfer 20 μl from well 1 to well 2 and repeat to produce the concentrations shown in **Image 6.1B**
4. Add 5 μl of a culture of the bacterial strain to wells 1–10.
5. Add 10 μl of culture medium, 10 μl of distilled water and 5 μl of a culture of the bacterial strain to well 12.
6. Repeat Steps 1–5 for two more rows in the microdilution plate.
7. Incubate plates at 37 $^{\circ}\text{C}$ for 24 hours.
8. Determine MIC by finding the lowest concentration at which no bacterial growth was seen in the well.

Answer the following questions about the experiment.

- (i) State the purpose of the culture medium.

[1]

- (ii) I. Use the volumes specified in **Image 6.1B** to describe how the concentration of antibiotics was produced in **well 1**.

[2]

- II. Name the method by which the different concentrations were produced.

[1]

- (iii) I. Explain why **well 12** was set up using the volumes stated in **step 5** of the method.

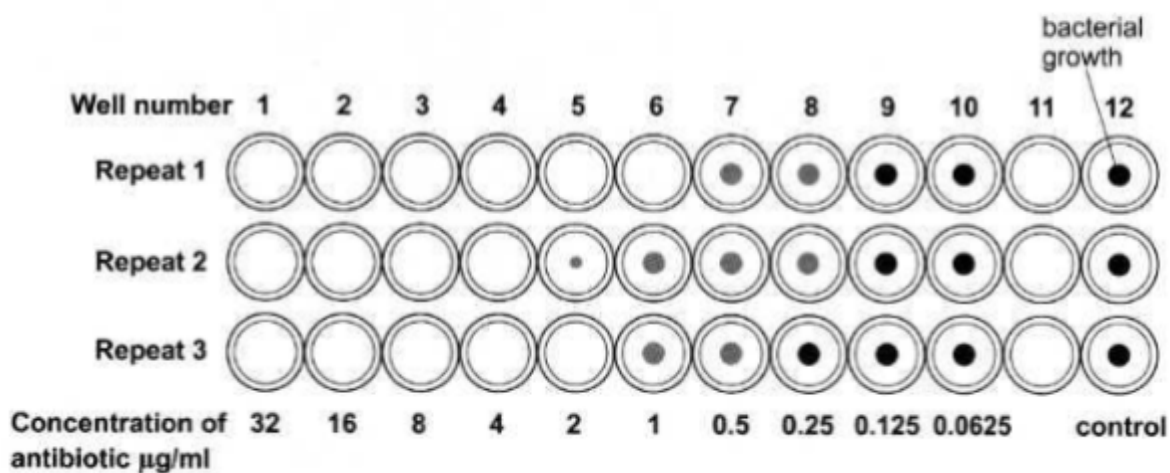
[1]

- II. Explain the purpose of **well 12** in this experiment

[1]

The method described on page 20 was used to determine the MIC of antibiotic C against E.coli and S.aureus. The results for E.coli are shown in **Image 6.2**

Image 6.2



(iv) I. For each of the repeats 1, 2, and 3, find the lowest concentration of antibiotic C that inhibits the growth of E. coli

[1]

Repeat 1 _____ µg / ml

Repeat 2 _____ µg / ml

Repeat 3 _____ µg / ml

II. Calculate the mean value for MIC in µg/ml.

[1]

Mean MIC = _____ µg/ml

(v) Describe **one** limitation of this method for determining MIC.

[1]

- (b) The method on Page 20 was also used to find the MIC for antibiotics A, B and D. The MIC for the four antibiotics against *E. coli* and *S. aureus* are shown in **Table 6.3**.

Table 6.3

Antibiotic	Minimum Inhibitory Concentration / μM	
	<i>E. coli</i>	<i>S. aureus</i>
A	16.13	>64.52
B	>67.01	>67.01
C	1.17	18.71
D	8.19	>65.50

- (i) With reference to the results in **Table 6.3**, state which antibiotic (A-D) shows most promise as a potential new treatment. Explain your answer.

[1]

Antibiotic _____

- (ii) *E. coli* is Gram negative, but *S. aureus* is Gram positive.

- I. Describe **one** difference between the cell walls of the two types of bacteria

[1]

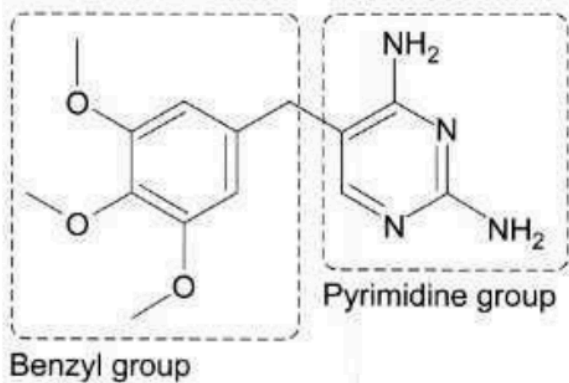
- II. Suggest how the difference in the cell walls might account for the MIC's shown in **Table 6.3** for antibiotic A, C and D.

[1]

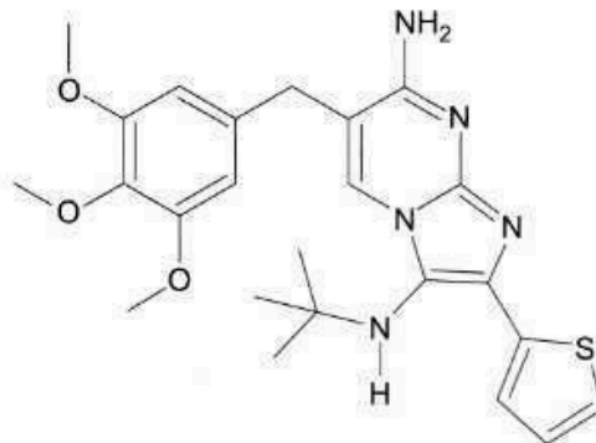
(c) Trimethoprim (TMP) is an antibiotic that affects DNA synthesis. Compound X has been identified as a possible new antibiotic. Image 6.4 shows the chemical structures of trimethoprim and compound X.

Image 6.4

Trimethoprim



Compound X



(i) Name one pyrimidine that could be found in a DNA nucleotide

[1]

(ii) TMP is a competitive inhibitor of dihydrofolate reductase, which is an enzyme involved in DNA synthesis.

With reference to **Image 6.4**, explain how compound X might also inhibit dihydrofolate reductase.

[2]

(iii) Compound X was extracted from a plant discovered growing in a tropical rainforest. Explain the importance of conserving plant species such as this.

[2]

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